

1. Suppose the monthly cost for the manufacture of golf balls is $C(x) = 3200 + 0.52x$ dollars, where x is the number of golf balls produced each month. Also, suppose the monthly revenue for the sale of golf balls is $R(x) = 1.32x$ dollars for x golfballs sold.

(A) The model for profit, including output and input units, would be:

$P(x) =$

☐ A. dollars

for x

☐ A. dollars

☐ B. golfballs

☐ B. golfballs

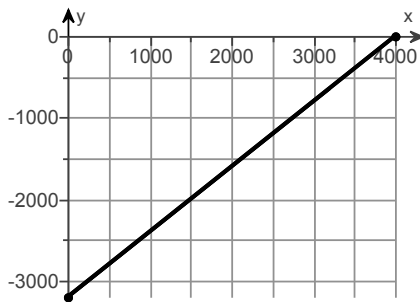
(B) The rate of change for profit is

☐ A. golfballs per

☐ A. dollar

☐ B. dollars

☐ B. golfball



y - intercept

x - intercept

(0, - 3200)

(4000,0)

(C) The y - intercept tells us:

- ☐ A. When we sell 0 golfballs we lose 3200 dollars.
☐ B. We have to sell 3200 golfballs to break even.
☐ C. When we sell 0 goldballs we get a 3200 dollar profit.

(C) The x - intercept tells us:

- ☐ A. We have to sell 4000 golfballs to break even.
☐ B. When we sell 0 goldballs we get a 4000 dollar profit.
☐ C. When we sell 0 golfballs we lose 3200 dollars.

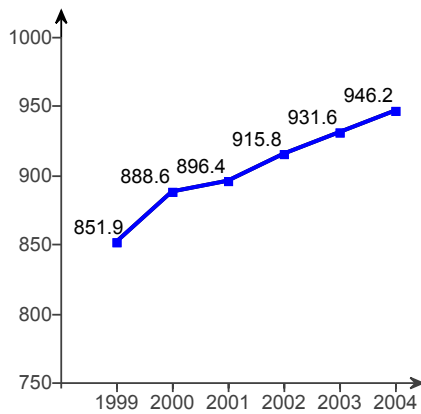
2. A company finds it can produce 25 heaters for \$7100, while producing 30 heaters costs \$8400. Express the cost, y , as a linear function of the number of heaters, x .

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$y =$

(Type an expression using x as the variable. Simplify your answer. Do not factor.)

- *3. The graph below gives the number of millions of persons who took trips of 50 miles or more for the years 1999 through 2004.



Answer parts (a) through (c).

- (a) Find the equation of the line which is the best fit for these data, with x equal to the number of years after 1999 and y equal to the millions of persons.

$y =$

(Use integers or decimals for any numbers in the expression. Round to three decimal places as needed.)

- (b) Use the model to estimate the number of persons who took trips in 2008.

millions

(Round to two decimal places as needed.)

- (c) In what year does the model estimate the number of persons who took trips as 985.72 million?

(Round to the nearest whole number as needed.)